CYBERSRS

Firewall Management System Requirements

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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1. Introduction

Original project description

Design a centralized firewall management system for a university that manages firewall policies across academic, administrative, research, hostel, and guest networks. The system should allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts. The platform should maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings. The solution should be secure, scalable, reliable, and manageable.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a centralized firewall management system for a university that manages firewall policies across academic, administrative, research, hostel, and guest networks.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: The system should allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: The platform should maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: The solution should be secure, scalable, reliable, and manageable.

Status: confirmed

Evidence ids: E-004

2. Functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

FR-001 - provide a centralized firewall management interface to manage firewall policies across academic, administrative, research, hostel, and guest networks.

Statement: The system shall provide a centralized firewall management interface to manage firewall policies across academic, administrative, research, hostel, and guest networks.

Rationale: Implements stakeholder source E-001.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: A network administrator with access privileges

When: The administrator initiates policy management for the academic network segment

Then: The system shall present a configuration interface for defining firewall policies specific to that segment

Method: test

Observable evidence: Test capture of the policy creation interface response

FR-002 - allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Statement: The system shall allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Rationale: Implements stakeholder source E-002.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: An administrator with policy creation privileges and a test environment configured

When: The administrator creates a new rule to restrict access from hostel to research networks

Then: The system shall enforce the rule and block unauthorized traffic between segments as defined

Method: test

Observable evidence: Test capture of traffic flow before and after rule enforcement

Given: An administrator with review privileges and a live network segment

When: The administrator requests to review existing firewall rules for the administrative network

Then: The system shall return the current set of active rules in a readable format

Method: test

Observable evidence: Test capture of rule response during review

Given: A network segment with abnormal traffic patterns

When: The system detects suspicious communication attempts between guest and research networks

Then: The system shall generate an alert indicating unauthorized communication

Method: test

Observable evidence: Test capture of the alert notification response

Given: An administrator with update privileges and a live policy in place

When: The administrator modifies a rule to allow access from academic to administrative networks

Then: The administrator modifies a rule to allow access from academic to administrative networks

Method: test

Observable evidence: Test capture of resulting policy state

FR-003 - The platform shall maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings.

Statement: The platform shall maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings.

Rationale: Implements stakeholder source E-003.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: An administrator performs a policy change on the research network segment

When: The action is completed successfully

Then: The system shall record the change in audit logs with timestamp, user ID, and action type

Method: test

Observable evidence: Test capture of audit log entry after policy modification

Given: A network administrator accesses the security dashboard

When: The dashboard is loaded for the academic building segment

Then: The dashboard is loaded for the academic building segment

Method: test

Observable evidence: Test capture of dashboard response

Given: A campus building with multiple network segments is configured

When: The administrator selects the hostel building in the interface

Then: The system shall present firewall policies and events specific to that building

Method: test

Observable evidence: Test capture of segment-specific policy view for hostel

3. Non-functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

NFR-001 - The solution shall be secure, scalable, reliable, and manageable.

Statement: The solution shall be secure, scalable, reliable, and manageable.

Rationale: Implements stakeholder source E-004.

Priority: should

Dimensions: availability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A system administrator evaluates the operational readiness of the network infrastructure

When: The administrator requests an assessment of security, scalability, reliability, and manageability attributes

Then: The administrator receives an assessment of security, scalability, reliability, and manageability attributes

Method: inspection

Observable evidence: Test capture of resulting assessment response

Unresolved decisions: Definition of scalability thresholds in terms of concurrent users or network traffic volume; Quantitative reliability metrics (e.g., uptime percentage) to be met during peak operations; Specific manageability criteria such as interface usability, configuration speed, and error recovery

4. Security, privacy and compliance

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-003

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003

7. Compliance and policy applicability matrix

Open decisions: Which specific compliance framework or policy is intended to be applied to the firewall management system?; What are the defined thresholds for scalability and reliability in terms of concurrent users, network traffic volume, and uptime percentage?; How will manageability criteria such as interface usability, configuration speed, and error recovery be measured and validated?

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Actor: authorized network administrators

Detail: Proposed logical endpoint for the source-stated actor: authorized network administrators.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002

Component name: Actor: authorized network administrators

Responsibility: Represent the source-stated actor: authorized network administrators.

Firewall Policy Management Interface

Detail: Proposed logical owner for FR-001, NFR-001.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-001; NFR-001

Component name: Firewall Policy Management Interface

Responsibility: The system shall provide a centralized firewall management interface to manage firewall policies across academic, administrative, research, hostel, and guest networks. The solution shall be secure, scalable, reliable, and manageable.

Rule Enforcement Engine

Detail: Proposed logical owner for FR-002, NFR-001.

Status: proposed

Evidence ids: E-002; E-004

Requirement ids: FR-002; NFR-001

Component name: Rule Enforcement Engine

Responsibility: The system shall allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts. The solution shall be secure, scalable, reliable, and manageable.

Audit and Monitoring Dashboard

Detail: Proposed logical owner for FR-003, NFR-001.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-003; NFR-001

Component name: Audit and Monitoring Dashboard

Responsibility: The platform shall maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings. The solution shall be secure, scalable, reliable, and manageable.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-002.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002

From component: Actor: authorized network administrators

To component: Rule Enforcement Engine

Data or action: The system shall allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-002.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002

Untrusted component: Actor: authorized network administrators

Trusted component: Rule Enforcement Engine

Enforcement: The system shall allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for NFR-001.

Status: proposed

Evidence ids: E-004

Requirement ids: NFR-001

Untrusted component: Firewall Policy Management Interface

Trusted component: Rule Enforcement Engine

Enforcement: The solution shall be secure, scalable, reliable, and manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for NFR-001.

Status: proposed

Evidence ids: E-004

Requirement ids: NFR-001

Untrusted component: Firewall Policy Management Interface

Trusted component: Audit and Monitoring Dashboard

Enforcement: The solution shall be secure, scalable, reliable, and manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for NFR-001.

Status: proposed

Evidence ids: E-004

Requirement ids: NFR-001

Untrusted component: Rule Enforcement Engine

Trusted component: Firewall Policy Management Interface

Enforcement: The solution shall be secure, scalable, reliable, and manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for NFR-001.

Status: proposed

Evidence ids: E-004

Requirement ids: NFR-001

Untrusted component: Rule Enforcement Engine

Trusted component: Audit and Monitoring Dashboard

Enforcement: The solution shall be secure, scalable, reliable, and manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for NFR-001.

Status: proposed

Evidence ids: E-004

Requirement ids: NFR-001

Untrusted component: Audit and Monitoring Dashboard

Trusted component: Firewall Policy Management Interface

Enforcement: The solution shall be secure, scalable, reliable, and manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for NFR-001.

Status: proposed

Evidence ids: E-004

Requirement ids: NFR-001

Untrusted component: Audit and Monitoring Dashboard

Trusted component: Rule Enforcement Engine

Enforcement: The solution shall be secure, scalable, reliable, and manageable.

9. Threats

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

11. User stories and use cases

User stories and use cases

Id: UC-001

Actor: authorized network administrators

Goal: The system shall allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Story: As authorized network administrators, I need the following behavior: The system shall allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Preconditions

1. An administrator with review privileges and a live network segment

Steps

1. The administrator requests to review existing firewall rules for the administrative network

Success outcome: The system shall return the current set of active rules in a readable format

Alternate flows

1. Given: An administrator with policy creation privileges and a test environment configured When: The administrator creates a new rule to restrict access from hostel to research networks Then: The system shall enforce the rule and block unauthorized traffic between segments as defined Evidence: Test capture of traffic flow before and after rule enforcement
2. Given: A network segment with abnormal traffic patterns When: The system detects suspicious communication attempts between guest and research networks Then: The system shall generate an alert indicating unauthorized communication Evidence: Test capture of the alert notification response

Requirement ids: FR-002

Evidence ids: E-002

Id: UC-002

Actor: authorized network administrators

Goal: The system shall allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Story: As authorized network administrators, I need the following behavior: The system shall allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Preconditions

1. An administrator with update privileges and a live policy in place

Steps

1. The administrator modifies a rule to allow access from academic to administrative networks

Success outcome: The administrator modifies a rule to allow access from academic to administrative networks

Requirement ids: FR-002

Evidence ids: E-002

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: A network administrator with access privileges
2. When: The administrator initiates policy management for the academic network segment

Expected result: Expected: The system shall present a configuration interface for defining firewall policies specific to that segment

Evidence to record: Evidence: Test capture of the policy creation interface response

Test type: positive

Id: TEST-002

Requirement ids: FR-002

Procedure

1. Given: An administrator with policy creation privileges and a test environment configured
2. When: The administrator creates a new rule to restrict access from hostel to research networks

Expected result: Expected: The system shall enforce the rule and block unauthorized traffic between segments as defined

Evidence to record: Evidence: Test capture of traffic flow before and after rule enforcement

Test type: positive

Id: TEST-003

Requirement ids: FR-002

Procedure

1. Given: An administrator with review privileges and a live network segment
2. When: The administrator requests to review existing firewall rules for the administrative network

Expected result: Expected: The system shall return the current set of active rules in a readable format

Evidence to record: Evidence: Test capture of rule response during review

Test type: positive

Id: TEST-004

Requirement ids: FR-002

Procedure

1. Given: A network segment with abnormal traffic patterns
2. When: The system detects suspicious communication attempts between guest and research networks

Expected result: Expected: The system shall generate an alert indicating unauthorized communication

Evidence to record: Evidence: Test capture of the alert notification response

Test type: positive

Id: TEST-005

Requirement ids: FR-002

Procedure

1. Given: An administrator with update privileges and a live policy in place
2. When: The administrator modifies a rule to allow access from academic to administrative networks

Expected result: Expected: The administrator modifies a rule to allow access from academic to administrative networks

Evidence to record: Evidence: Test capture of resulting policy state

Test type: positive

Id: TEST-006

Requirement ids: FR-003

Procedure

1. Given: An administrator performs a policy change on the research network segment
2. When: The action is completed successfully

Expected result: Expected: The system shall record the change in audit logs with timestamp, user ID, and action type

Evidence to record: Evidence: Test capture of audit log entry after policy modification

Test type: positive

Id: TEST-007

Requirement ids: FR-003

Procedure

1. Given: A network administrator accesses the security dashboard
2. When: The dashboard is loaded for the academic building segment

Expected result: Expected: The dashboard is loaded for the academic building segment

Evidence to record: Evidence: Test capture of dashboard response

Test type: positive

Id: TEST-008

Requirement ids: FR-003

Procedure

1. Given: A campus building with multiple network segments is configured
2. When: The administrator selects the hostel building in the interface

Expected result: Expected: The system shall present firewall policies and events specific to that building

Evidence to record: Evidence: Test capture of segment-specific policy view for hostel

Test type: positive

Id: TEST-009

Requirement ids: NFR-001

Procedure

1. Given: A system administrator evaluates the operational readiness of the network infrastructure
2. When: The administrator requests an assessment of security, scalability, reliability, and manageability attributes

Expected result: Expected: The administrator receives an assessment of security, scalability, reliability, and manageability attributes

Evidence to record: Evidence: Test capture of resulting assessment response

Test type: inspection

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Policy

Detail: A rule or decision governing permitted actions; its project-specific meaning remains defined by the source requirements.

Status: proposed

Evidence ids: E-001; E-003

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-003

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-003

Firewall

Detail: A control that permits or restricts network traffic according to rules; mentioning it does not authorize automatic changes.

Status: proposed

Evidence ids: E-001; E-002; E-003

14. SDLC recommendation and workflow

Selected model: Agile

Justification: No differentiating input evidence is provided to select among Agile, DevSecOps, Spiral, V-Model, Waterfall, Hybrid, or Incremental models. All candidates receive equal suitability points (40) due to absence of factors such as budget, delivery, failure consequences, feedback, legacy, regulation, schedule, stability, uncertainty, verification. The selection is therefore provisional and subject to later evidence.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall provide a centralized firewall management interface to manage firewall policies across academic, administrative, research, hostel, and guest networks.; FR-002: The system shall allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.; FR-003: The platform shall maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings.
2. Review and baseline the source-defined work group: NFR-001: The solution shall be secure, scalable, reliable, and manageable.
3. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline

2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001

Name: Agile delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall provide a centralized firewall management interface to manage firewall policies across academic, administrative, research, hostel, and guest networks.; FR-002: The system shall allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.; FR-003: The platform shall maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings.
3. Implement the source-defined work group: NFR-001: The solution shall be secure, scalable, reliable, and manageable.
4. In an authorized isolated test environment, execute TEST-001: Given: A network administrator with access privileges When: The administrator initiates policy management for the academic network segment Expected: The system shall present a configuration interface for defining firewall policies specific to that segment
5. In an authorized isolated test environment, execute TEST-002: Given: An administrator with policy creation privileges and a test environment configured When: The administrator creates a new rule to restrict access from hostel to research networks Expected: The system shall enforce the rule and block unauthorized traffic between segments as defined
6. In an authorized isolated test environment, execute TEST-003: Given: An administrator with review privileges and a live network segment When: The administrator requests to review existing firewall rules for the administrative network Expected: The system shall return the current set of active rules in a readable format
7. In an authorized isolated test environment, execute TEST-004: Given: A network segment with abnormal traffic patterns When: The system detects suspicious communication attempts between guest and research networks Expected: The system shall generate an alert indicating unauthorized communication
8. In an authorized isolated test environment, execute TEST-005: Given: An administrator with update privileges and a live policy in place When: The administrator modifies a rule to allow access from academic to administrative networks Expected: The administrator modifies a rule to

allow access from academic to administrative networks

9. In an authorized isolated test environment, execute TEST-006: Given: An administrator performs a policy change on the research network segment When: The action is completed successfully Expected: The system shall record the change in audit logs with timestamp, user ID, and action type
10. In an authorized isolated test environment, execute TEST-007: Given: A network administrator accesses the security dashboard When: The dashboard is loaded for the academic building segment Expected: The dashboard is loaded for the academic building segment
11. In an authorized isolated test environment, execute TEST-008: Given: A campus building with multiple network segments is configured When: The administrator selects the hostel building in the interface Expected: The system shall present firewall policies and events specific to that building
12. In an authorized isolated test environment, execute TEST-009: Given: A system administrator evaluates the operational readiness of the network infrastructure When: The administrator requests an assessment of security, scalability, reliability, and manageability attributes Expected: The administrator receives an assessment of security, scalability, reliability, and manageability attributes
13. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
14. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-1.1

Generation profile: sequential-sdlc-generation-2.0

Ranked candidates

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: DevSecOps

Score: 40

Reasons: No differentiating input evidence

Model: Spiral

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Model: Hybrid

Score: 40

Reasons: No differentiating input evidence

Model: Incremental

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a centralized firewall management system for a university that manages firewall policies across academic, administrative, research, hostel, and guest networks.

Id: E-002

Source: description

Quote: The system should allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Id: E-003

Source: description

Quote: The platform should maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings.

Id: E-004

Source: description

Quote: The solution should be secure, scalable, reliable, and manageable.

Source references

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7085

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.7132

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-788cd8fba7ce0acb:3 - 03_OWASP_CheatSheets/083_OWASP_Network_Segmentation_Cheat_Sheet.md

Source document id: kb-788cd8fba7ce0acb

Chunk index: 3

Page or section: text chunk 4

Relevance score: 0.7096

Sha256: 2b91c4e103da8e53c98e3b6006b23c2bb02e8c55e5805a56f0068c7b31a477d8

kb-c1e0239af9dd46bd:14 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 14

Page or section: text chunk 15

Relevance score: 0.7034

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kb-c1e0239af9dd46bd:9 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 9

Page or section: text chunk 10

Relevance score: 0.7019

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-788cd8fba7ce0acb:4 - 03_OWASP_CheatSheets/083_OWASP_Network_Segmentation_Cheat_Sheet.md

Source document id: kb-788cd8fba7ce0acb

Chunk index: 4

Page or section: text chunk 5

Relevance score: 0.7078

Sha256: 2b91c4e103da8e53c98e3b6006b23c2bb02e8c55e5805a56f0068c7b31a477d8

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: qwen3:4b-instruct-2507-q4_K_M

Non-functional requirements: qwen3:4b-instruct-2507-q4_K_M

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: cybersrs-qwen3-4b-ft:latest

Prompt version: sequential-1.30

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	TEST-001, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-002	E-002	UC-001, UC-002, TEST-002, TEST-003, TEST-004, TEST-005, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-003	E-003	TEST-006, TEST-007, TEST-008, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-001	E-004	TEST-009, Requirements and verification planning, Agile delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.30